

PROFESSIONAL SYLLABUS

Fundamentals & Investigation Techniques

Master the science of identifying, preserving, and analyzing digital evidence — across computers, networks, memory, mobile, and cloud environments — with industry-standard tools, legal frameworks, and a real-world investigation capstone.

MODULES

11 Modules

LEVEL

Beginner → Pro

FORMAT

Labs + Capstone

OUTCOME

DFIR-Ready



obscyratech@gmail.com



+91 96062 62858 | +91 91104
60406



Mysore, Karnataka,
India

Course Description

Digital Forensics is the science of identifying, preserving, collecting, analyzing, and presenting digital evidence in a **legally defensible manner**. This course provides students with the knowledge and practical skills required to conduct digital investigations involving computers, networks, mobile devices, cloud environments, and emerging technologies.

The course combines forensic theory, legal principles, investigative methodologies, and **hands-on laboratory exercises** to prepare learners for careers in **digital forensics, incident response, cybercrime investigations, law enforcement support, and cybersecurity operations**.

REQUIRED PREREQUISITES

- ✓ Basic Computer Knowledge
- ✓ Basic Operating System Knowledge

RECOMMENDED PREREQUISITES

- ★ Cybersecurity Fundamentals
- ★ Network Security Fundamentals

Learning Objectives

- ✓ Understand the digital forensic investigation lifecycle.
- ✓ Acquire and preserve digital evidence using industry-standard methodologies.
- ✓ Analyze file systems and recover deleted data.
- ✓ Perform memory and volatile data analysis.
- ✓ Investigate network activity and log data.
- ✓ Conduct basic mobile and cloud forensic investigations.
- ✓ Identify malware artifacts and indicators of compromise.
- ✓ Maintain chain of custody and follow legal requirements.
- ✓ Prepare professional forensic reports and present findings.
- ✓ Evaluate modern forensic challenges including anti-forensics, IoT, and AI-generated evidence.

Syllabus Index

CORE CURRICULUM		
01	Introduction to Digital Forensics	03
02	Legal, Ethical & Regulatory Considerations	03
03	Digital Evidence Acquisition & Preservation	04
04	File System Forensics	04
05	Data Recovery & Artifact Analysis	05
06	Windows Forensics	05
07	Linux & macOS Forensics	06
08	Memory & Volatile Data Forensics	06
09	Network Forensics	07
10	Log Analysis & Security Events	07
11	Malware & Artifact Analysis	08
PROGRAM ADD-ONS		
★	Required Tools & Software	09
★	Assessment & Grading	09
★	Recommended Readings & References	10
★	Final Forensic Investigation Project	10
★	Career Pathways	11

Course Modules

MODULE 01 Introduction to Digital Forensics

TOPICS

- What is Digital Forensics?
- Digital Evidence Fundamentals
- Investigation Lifecycle
- Types of Digital Crimes
- Forensic Readiness

LAB Identify and classify digital evidence from sample scenarios.

MODULE 02 Legal, Ethical & Regulatory Considerations

TOPICS

- Cybercrime Laws
- Privacy Considerations
- Search and Seizure
- Evidence Admissibility
- Chain of Custody

LAB Prepare a chain-of-custody document.

MODULE 03 Digital Evidence Acquisition & Preservation

TOPICS

- Evidence Collection
- Forensic Imaging
- Hashing and Verification
- Live Acquisition
- Static Acquisition

LAB Create and verify a forensic image using FTK Imager.

MODULE 04 File System Forensics**TOPICS**

- FAT File System
- NTFS File System
- ext4 File System
- APFS Overview
- File Metadata

LAB Analyze file system structures and metadata.

MODULE 05 Data Recovery & Artifact Analysis**TOPICS**

- Deleted File Recovery
- Recycle Bin Analysis
- Browser Artifacts
- User Activity Analysis
- Metadata Investigation

LAB Recover deleted files from forensic images.

MODULE 06 Windows Forensics**TOPICS**

- Windows Registry
- Event Logs
- User Accounts
- Prefetch Files
- Windows Artifacts

LAB Investigate a Windows workstation image.

MODULE 07 Linux & macOS Forensics**TOPICS**

- Linux Logs
- User Activity
- Shell History
- macOS Artifacts
- System Configuration Files

LAB Analyze Linux forensic evidence.

MODULE 08 Memory & Volatile Data Forensics**TOPICS**

- RAM Acquisition
- Process Analysis
- Running Services
- Network Connections
- Malware Indicators

LAB Analyze memory dumps using Volatility.

MODULE 09 Network Forensics**TOPICS**

- Packet Analysis
- Network Traffic Investigation
- Intrusion Evidence
- DNS Analysis
- Email Forensics

LAB Investigate network captures using Wireshark.

MODULE 10 Log Analysis & Security Events**TOPICS**

- System Logs
- Authentication Logs
- Application Logs
- Security Event Correlation
- Timeline Analysis

LAB Build an event timeline from log sources.

MODULE 11 Malware & Artifact Analysis**TOPICS**

- Malware Investigation Basics
- Persistence Mechanisms
- Indicators of Compromise
- Behavioral Analysis
- Malware Artifacts

LAB Analyze malware indicators in a controlled environment.

Required Tools & Software

Industry-standard digital forensic tools used throughout the program.

FORENSIC SUITES

- ▶ Autopsy
- ▶ FTK Imager
- ▶ EnCase Forensic

MEMORY ANALYSIS

- ▶ Volatility
- ▶ Rekall

NETWORK ANALYSIS

- ▶ Wireshark
- ▶ NetworkMiner

MOBILE FORENSICS

- ▶ MOBILedit Forensic
- ▶ ADB (Android Debug Bridge)

SUPPORTING TOOLS

- ▶ VirtualBox
- ▶ HxD (Hex Editor)
- ▶ ExifTool

LAB ENVIRONMENT

- ▶ Isolated VM-based lab
- ▶ Sample forensic images
- ▶ Pre-built case scenarios

Assignments & Assessment

Balanced evaluation across labs, assignments, quizzes, midterm, case studies, and a final investigation project.

ASSESSMENT COMPONENT	WEIGHTAGE
Weekly Labs	25%
Practical Assignments	20%
Total	100%

ASSESSMENT COMPONENT	WEIGHTAGE
Quizzes	10%
Midterm Examination	15%
Case Study Analysis	10%
Final Forensic Investigation Project	20%
Total	100%

Recommended Readings & References

Curated textbooks, official standards, and additional learning resources.

▢ Textbooks

- Guide to Computer Forensics and Investigations
- Digital Forensics with Open Source Tools
- Practical Digital Forensics

▢ Standards & Guidelines

- NIST Digital Forensics Publications
- ISO/IEC 27037
- SANS Institute Forensics Resources

▢ Additional Resources

- DFIR Training Materials
- Incident Response Case Studies
- Open-Source Forensic Challenges
- Digital Evidence Handling Guides

▢ Practice Platforms

- CyberDefenders Blue Team Labs
- TryHackMe DFIR Paths
- DFIR.training Test Images
- Honeynet Project Challenges

Final Forensic Investigation Project

CAPSTONE PROJECT

Digital Forensic Investigation Case

Students will conduct a complete forensic investigation on a simulated incident.

PROJECT DELIVERABLES

- ✓ Evidence Acquisition Plan
- ✓ Chain of Custody Documentation
- ✓ File System Analysis
- ✓ Memory Analysis Findings

ADDITIONAL OUTPUTS

- ✓ Network Investigation Results
- ✓ Timeline Reconstruction
- ✓ Final Forensic Report
- ✓ Presentation of Findings

Expected Outcome: Students will demonstrate the ability to perform a **legally defensible forensic investigation** from evidence acquisition through reporting — applying forensic methodologies, legal principles, and analytical techniques learned throughout the course.

Career Pathways After This Course

Roles in the digital forensics, incident response, and cybercrime investigation industries:

Forensics & DFIR Roles

- Digital Forensics Analyst
- DFIR Analyst
- Incident Response Analyst
- Malware Analyst (Entry Level)

Law Enforcement & Investigation

- Cybercrime Investigator
- Law Enforcement Digital Examiner
- Fraud & Financial Crime Analyst
- Forensic Consultant

Cybersecurity Operations

- SOC Analyst (Tier 1 / Tier 2)
- Threat Hunter
- Security Engineer (Forensics Focus)
- Blue Team Analyst

Specialized Roles

- eDiscovery Specialist
- Cloud Forensics Analyst
- Mobile Forensics Examiner
- Memory / Reverse Engineering Analyst

Recommended advanced learning tracks at Obscyratech:

Advanced Malware Analysis

Threat Hunting

Cloud Forensics

Mobile Forensics Advanced

Incident Response Management

Reverse Engineering

Program Promise: This Digital Forensics Fundamentals program is mapped to globally recognized standards (**NIST, ISO/IEC 27037, SANS DFIR**) and prepares learners for careers in **Digital Forensics, DFIR, Incident Response, Cybercrime Investigation, and SOC operations**. Every module blends theory, legal principles, and hands-on labs — culminating in a complete capstone forensic investigation.

OBSCYRATECH

CYBERSECURITY • TRAINING • SERVICES

Follow the evidence. Solve the case.

Join Obscyratech's Digital Forensics Fundamentals program and learn how to acquire, analyze, and present digital evidence with the same methodologies trusted by global DFIR teams, law enforcement agencies, and cybersecurity professionals.

GET IN TOUCH

EMAIL

obs cyratech@gmail.com

PHONE

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+91 91104 60406

LOCATION

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PROGRAM

**Digital Forensics
Fundamentals**